

MID-SEMESTER EXAMINATION, April-2025 **Computer Networking: Security (CSE 3752)**

Program: B.Tech (CSE/CSIT)
Full Marks: 30

Semester: 6th
Time: 2 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the various types of security controls and fundamental security concepts.	L1	1	6
Explain the importance of using appropriate cryptographic solutions like PKI, encryption levels, tools, obfuscation, hashing, digital signatures, and certificates.	L3	2,3,4	18
Explain common threat actors, their motivations, and various threat vectors, along with understanding different types of vulnerabilities.	L2	5	6
Analyze the indicators of malicious activity, based on a given scenario and Explain the purpose of mitigation techniques used to secure the enterprise.			
Explain the importance of the security implications of different architecture models, including cloud, serverless, microservices, and network infrastructure			
Apply security principles in different infrastructure scenarios including device placement, security zones, and network appliances			

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

- (a) Briefly explain various control types with respect to security concern that help organizations to carry out the operations smoothly at their end. 2

(b) Justify that, the three principles of CIA triad when works in harmony create a robust defense against cyber threats. 2

- (c) Briefly highlight the importance of the following concepts associated with the change management process for mitigating the potential disaster effect. 2

(i) Impact analysis (ii) Test results (iii) Back out plan
(iv) Maintenance window.

2. (a) With the help of suitable diagram differentiate the operation of symmetric and asymmetric encryption technique. 2

- (b) What is meant by public and private key associated with public key infrastructure? Distinguish the use of both for the applications (i) Confidentiality (ii) Authentication. 2

- (c) Using the keyword "EDUCATION", construct the 5 X 5 matrix key for play fair ciphering and compute the cipher text for the plaintext "HELLO WORLD". Assume the letters I & J share the same cell in key matrix. 2

3. (a) State, how DES algorithm gives a better security as compared to the classical symmetric cryptographic algorithms. 2

- (b) Given a 8-bit input as (3A)_{Hex} to the following permutation table. 2

4	7	1	2
6	8	3	5

Find: (i) Respective 8 bit permuted output(in Hex form)

(ii) Inverse permutation table

- (c) How many S boxes are used in DES? Given the elements of the 2nd row (i.e. row no. "01") in a S box used in DES as 14,4,13,1,2,15,11,8,3,10,6,12,5,9,0 & 7. If the 6 bits input to the S box is "011011", Determine the corresponding 4 bit output? 2

4. (a) Given the plaintext "MEETING POSTPONED" to AES 128 bit algorithm. If the 128 bit key used for encryption as (02020202020202020202020202020202)Hex, then 2

(a) Show the original contents of state array, displayed as a 4 X 4 matrix.

(b) Show the value of state array after initial Add Round Key transformation.

- (b) State the use of RSA algorithm in data communication following cryptography process. Using RSA algorithm, find the value of 'e' if $p=3$, $q=11$ and $d=7$. 2

- (c) A pair of sender and receiver want to use the Diffie-Hellman algorithm to securely share a secret key. They agree to have the prime number and primitive root as 23 and 7 respectively. If the private key used by the sender is 5 and the receiver is 3, then compute : 2

(i) The data they transmitted to each other for sharing the secret key.

(ii) The value of shared key received by each of them. Also, justify that both of them has received the same key.

5. (a) An individual without a lot of experience in IT launches a cyber attack, using readily available tools to disrupt a local government website temporarily. Define threat actor and identify the category in which this scenario best fits. 2

- (b) Employees in a company start using a cloud storage service without authorization, bypassing official IT protocols. Explain the term that best describes this situation. 2

- (c) State the difference between spear phishing from regular phishing. 2

End of Questions

MID-SEMESTER EXAMINATION, April-2025 **Essential Mathematics for Artificial Intelligence 2 (CSE 3788)**

Programme: B.Tech(CSE & CSIT)
Full Marks: 30

Semester: 6th
Time: 2 Hours

move to Bad state. If the company is in Bad state today, there is 50% chance that it will remain in Bad state and 50% chance that it will move to Good state. Derive the transition matrix for this system. Also calculate the probability that if today the company is in Good state, then after two days it will be in Good state.

- (c) Suppose there are two bowls of cookies. Bowl 1 contains 30 vanilla cookies and 10 chocolate cookies. Bowl 2 contains 20 of each. Now suppose you choose one of the bowls at random and, without looking, select a cookie at random. The cookie is vanilla. Calculate the probability that it came from Bowl 1.

End of Questions

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Able to understand about different types of probabilistic generative model and use of it in probabilistic language modelling.	L1, L3, L3, L6	1(a,b,c) 2(a)	2*4=8
Able to apply the knowledge of graph theory for solving different AI model.	L6, L3, L3	2(b,c) 3(a)	2*3=6
Able to apply machine learning into the field of operations research for more efficient and informed decision making.	L6, L3, L3, L3, L3	3(b,c) 4(a,b,c) 5(a)	2*6=12
Able to Analyse the use of different stochastic method for solving AI model.	L3, L3	5(b,c)	2*2=4
Able to Use the concept of mathematical logic and partial differential equation in the field of AI.			
Able to Understand about AI ethics and AI policy.			

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

- (a) What is probabilistic generative model? Give two example. 2
- (b) Let P and Q be two distributions on a random variable X which takes the values {0,1,2}. P is a Binomial distribution with $n=2$ and $p=0.5$ and Q is a uniform distribution on {0,1,2}. Calculate the KL-divergence of Q over P. 2

(c) Consider the data of car stolen and the car type:

Serial Number	Color	Type	Design	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	Yes
3	Red	Sports	Domestic	No
4	Yellow	Sports	Domestic	Yes
5	Yellow	Sports	Domestic	No
6	Yellow	SUV	Imported	Yes
7	Yellow	SUV	Imported	No
8	Yellow	SUV	Yes	Yes
9	Red	SUV	Domestic	No
10	Red	SUV	Imported	No
		Sports	Imported	Yes

Use Naïve Bayes Classification to predict whether the car will be stolen or not if the car is "Red, SUV, Domestic".

2. (a) Consider a random variable X whose probability distribution is

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

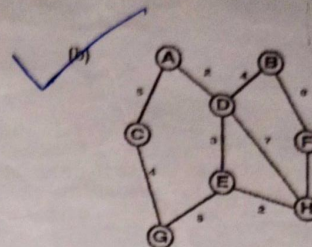
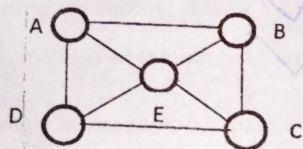
Derive the transformed distribution if the transformation function is $y(x) = -\frac{1}{\lambda} \ln x, (\lambda > 0)$.

- (b) Consider a web page with four networks: A, B, C, D. The link structure is as follows: A is linked to B and C, B is linked to C, C is linked to A, D is linked to A and C. Create the graph and transition probability matrix for this.

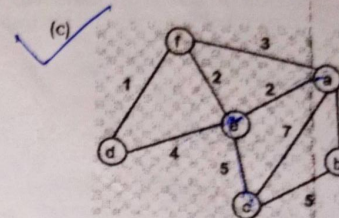
- (c) For the above network, calculate the Page Rank after three iteration.

3. (a) Consider the information vector $A[1,0,2]$, $B[1,2,4]$, $C[1,3,7]$, $D[2,4,8]$, $E[6,9,12]$ are associated as following network graph. Calculate the updated nodes after message passing when

$$\text{Aggregated message}(\text{node}) = \frac{\text{Product of node values}}{\text{Sum of node values}}$$



Considering the starting vertex as A, Construct the minimum spanning tree using Prim's algorithm.



Starting at node 'a', apply the Nearest Neighbour Algorithm to compute the path and total cost of this Travelling Salesman Problem.

4. (a) Solve the following Linear Programming Problem using Simplex method up to table 1 (Initial table and 1st table)

$$\text{Max } z = 4x_1 + 6x_2$$

$$\text{Subject to } 3x_1 + 2x_2 \leq 18, x_1 + 3x_2 \leq 12, x_1, x_2 \geq 0.$$

- (b) Solve the table 2 for the above linear programming problem, calculate the optimum solution and maximum value of Z .

- (c) Solve the above LPP by Graphical method.

5. (a) Calculate the maximum values of the function $f(x,y) = 2x^2 + 3y$ subject to the constraint

$$g(x,y) = x^2 + y^2 - 18 = 0 \text{ using Lagrangian multiplier method.}$$

- (b) A company has two states: Good(G): The company is profitable and Bad(B): The company is making a loss. If the company is in Good state today there is 80% chance it will remain in Good state tomorrow and 20% chance that it will

MID-SEMESTER EXAMINATION, April-2025 **Compilers: Principles, Techniques, and Tool (CSE 3739)**

Program: B.Tech (CSE, CSIT)
Full Marks: 30

Semester: 6th
Time: 2 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the overview of programming languages, language processors and the structure of a compiler.	L2,L3 L1,L2 L2,L3	1.a 1.b 1.c	6
Acquire knowledge in theory of computation and their role in designing different types of tokens generated by lexical analyzer.	L3 L3,L4 L1,L2	2.a 2.b 2.c	6
Understand the role of Parser(s) (LL, SLR, CLR and LALR) and its types i.e. Top-down and Bottom-up parsers.	L3,L4 L4,L5 L3,L4 L3,L4 L4,L5 L4,L5	3.a 3.b 3.c 4.a 4.b,c 5.a,b, c	18

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1.	(a)	Consider the following fragment of 'C' code: <pre>float a,b; a = a * -50 + b + 3;</pre> Write the output of 1 st Four phases of the compiler for the above arithmetic expression in the 'C' code.	2
	(b)	Write short notes on: i) Symbol table ii) Input buffering	2
	(c)	Explain the language processing system for the modern computer programming languages.	2

Write the regular expression for the floating point number with exponent. The example of some floating point numbers with exponent are some strings such as 6.5897E4, 1.8569E-4, 2.54e15 or 3.54e-15. [Hint: At least one digit should be before and after the dot (.)]

(b) Construct a transition diagram (finite state automaton) that represents the lexical recognition of a floating-point number with an exponent.

(c) Find the number of tokens in the given C statement.

`printf("pt = %d, &pt = %x", pt, &pt);`

[GATE 2000]

3. (a) Consider two grammars G with the production rules given below:

$G: S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$

$E \rightarrow b$

where if, then, else, a, b, c are the terminals.

Determine whether the given grammar G is ambiguous.

[GATE 2025]

(b) Determine whether the given grammar G in Question 3(a) is LL(1) grammar.

[GATE 2025]

(c) Consider the following context-free grammar where the start symbol is S and the set of terminals is {a, b, c, d}.

$S \rightarrow AaAb \mid BbBa$

$A \rightarrow cS \mid \epsilon$

$B \rightarrow dS \mid \epsilon$

The following is a partially-filled LL(1) parsing table.

	a	b	c	d	\$
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$	(1)	(2)	
A	$A \rightarrow \epsilon$	(3)	$A \rightarrow cS$		
B	(4)	$B \rightarrow \epsilon$		$B \rightarrow dS$	

Write down the CORRECT productions for the numbered cells in the parsing table?

[GATE 2024]

4. (a) Compute the FIRST() and FOLLOW() of the given grammar:

$S \rightarrow Aba \mid bCa$

$A \rightarrow cBcB \mid \epsilon$

$B \rightarrow CdA \mid ad$

$C \rightarrow eC \mid \epsilon$

$D \rightarrow bsf \mid a$

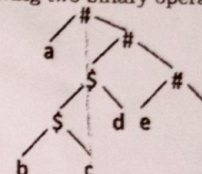
(b) The attributes of three arithmetic operators in some programming language are given below.

perator	Precedence	Associativity	Arity
+	High	Left	Binary
-	Medium	Right	Binary
*	Low	Left	Binary

Compute the value of the expression $2 - 5 + 1 - 7 * 3$ and create the parse tree for this expression.

[GATE 2016]

(c) Consider the following parse tree for the expression $a\#b\$c\$d\#e\#f$, involving two binary operators \$ and #.



Determine which operator has the high precedence and what is the associativity of the two operators \$ and #.

[GATE 2018]

5. (a) Consider the following grammar with production rules as :

$E \rightarrow T + E \mid T$

$T \rightarrow id$

Compute the canonical collection of LR(0) items of the grammar.

(b) Justify that the grammar given in question 5(a) is NOT LR(0) grammar.

(c) Construct the SLR(1) parsing table for the grammar given in question 5(a).

End of Questions

MID-SEMESTER EXAMINATION, April-2025
Introduction To Databases (CSE-3151)

Program: B.Tech (CSE & CSIT)
Full Marks: 30

Semester: 6th
Time: 2 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
To identify and explain the different components and functionalities of DBMS and their interdependence through the database architecture.	L1, L2	1a, 1b, 1c, 2a	8
To analyze an enterprise schema for given user requirements and apply the conceptual database design principles through ER modelling to construct the ER diagram and corresponding relation schemas.	L4	2b, 2c, 3a, 3b, 3c	10
To analyze and design relational database schema using decomposition and normalization techniques.	L5, L6	4a, 4b, 4c, 5a, 5b, 5c	12

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) List four significant differences between a file-processing system and DBMS. 2
- (b) Explain the differences between 2-tier and 3-tier architectures. Which is better suited for web applications? 2
- (c) What is the need of data abstraction in a database system? Further explain different levels of abstraction. 2
2. (a) What are the different types of database users? How do their roles impact the overall database system? 2
- (b) Explain simple, composite, multivalued and derived attributes with suitable examples. 2

- (c) What is the difference between overlapping and disjoint specialization? Develop the appropriate relational schemas for the given diagram in figure 1.

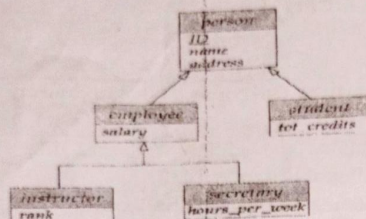


Figure 1.

3. (a) Consider a database maintaining the details of the orders placed by customers in an online shopping system. Each customer is represented by a unique id, name, phoneno, address. Customers may have more than one phoneno, and address includes the city, state and pincode. One customer can place many orders. Each order is represented by a unique order_no, amount, date and order_status. Order includes some products. Each product is represented by a unique product_no, product_name, description and price. The association of customer and order is represented by relationship named as "places" and association of order and product is represented by the relationship name "includes". One order can include many products and many orders can be placed for the same product. The database also keeps information about the customers who have registered but have not placed any order. Construct the ER diagram for the above database representing entity set, relationship set, mapping cardinality and participation constraints.
- (b) Develop the appropriate relational schemas for the database of online shopping system illustrated in the ER diagram related to que. 3(a).
- (c) Analyze the referential integrity for the relational schema resulted in the que. 3(b) and construct its corresponding schema diagram.
4. (a) Consider the following set F of functional dependencies on the relational schema $R(A, B, C, D, E, F)$, $F = \{A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A\}$. Compute the closure of B. Prove using Armstrong's axiom that AF is a super key.

- (b) Consider the following table T, and the set of functional dependencies F: $\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

A	B	C	D
a1	b1	c1	d1
a1	b1	c2	d2
a2	b1	c1	d3
a2	b1	c3	d4

Given the following records or tuples, indicate which record can be added in to table T without violating any of the functional dependencies in F. If a record or tuple cannot be legally added, justify it indicating which functional dependency is violated. Records: $\{a2, b1, c4, d8\}$, $\{a1, b1, c2, d5\}$, $\{a3, b1, c4, d3\}$, $\{a2, b2, c1, d8\}$.

- (c) Consider two sets of FDs, F and G, $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ and $G = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$. Are F and G equivalent? Find Canonical cover of F.

5. (a) Consider a schema $R(A, B, C, D)$ with the functional dependencies on the relation schema: $F = \{A \rightarrow B \text{ and } C \rightarrow D\}$. Check both the properties of decomposition if R is decomposed into $R1(A, B)$ and $R2(C, D)$. If any of properties are not satisfied then what should be the actual decomposition of the relation schema R.
- (b) Consider a relational schema $\text{FinalyearProject}(\text{student}, \text{researchTopic}, \text{consultationDay}, \text{supervisor})$ with the functional dependencies: $F = \{\text{student} \rightarrow \text{researchTopic}, \text{student}, \text{researchTopic} \rightarrow \text{supervisor}, \text{researchTopic}, \text{supervisor} \rightarrow \text{consultationDay}\}$. Which is the weakest normal form that the schema satisfies? Comment whether it satisfies the property of 3NF or not with reason. If no, redesign the schema satisfying the properties of 3NF. (GATE 2019)
- (c) Consider the relational schema $\text{StudentPerformance}(\text{Name}, \text{CourseNo}, \text{RegNo}, \text{Grade})$ has the following functional dependencies: $F = \{\text{Name}, \text{CourseNo} \rightarrow \text{Grade}, \text{RegNo}, \text{CourseNo} \rightarrow \text{Grade}, \text{Name} \rightarrow \text{RegNo}, \text{RegNo} \rightarrow \text{Name}\}$. Which is the highest normal form of this relation schema satisfies? (GATE 2004)

End of Questions

MID-SEMESTER EXAMINATION, April-2025

Python for Computer Science and Data Science 2 (CSE 3652)

Programme: B.Tech (CSE)
Full Marks: 30

Semester: 6th
Time: 2 Hours

4. (a) What is Cognitive Computing? Explain how IBM Watson Cloud leverages cognitive computing to enhance its capabilities. 2
- (b) Write a Python program to fetch user details, including the follower count, from Twitter's API using the ***tweepy*** package and authentication credentials. 2
- (c) Write a Python code snippet to implement the ***K-NN (K-Nearest Neighbors)*** algorithm and split the data into training and testing sets, with 70% for training and 30% for testing. Ensure reproducibility by setting the `random_state` parameter to 42. 2
5. (a) What is the difference between supervised and unsupervised machine learning? Provide examples of algorithms used in each type, and explain when each approach is typically used. 2
- (b) Define simple linear regression and provide its mathematical equation. Furthermore, if both training and test errors are high in a regression model, what is the likely cause, and how can it be addressed? 2
- (c) Consider the following confusion matrix for a binary classification problem. Compute the Accuracy, Precision, Recall, and F1-score: 2

Actual\Predicted	Positive (1)	Negative (0)
Positive (1)	TP (40)	FN (10)
Negative (0)	FP (5)	TN (45)

Where TP, FN, FP, and TN represent True Positives, False Negatives, False Positives, and True Negatives, respectively.

End of Questions

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Apply object-oriented programming principles and advanced Python features to design and test reusable, efficient applications for real-world problems.	L1, L2, L3	1 (a), 1(b), 1(c)	6
Analyze and implement recursive algorithms, searching, and sorting techniques, and evaluate their efficiency using Big O notation to optimize performance.	L3, L4	2 (a), 2 (b), 2 (c)	6
Apply natural language processing (NLP) techniques using tools and libraries to analyze, process, and visualize text data, including sentiment analysis, language translation, and entity recognition, with applications in machine learning.	L3, L4	3 (a), 3 (b), 3 (c)	6
Utilize APIs to collect, analyze, and visualize social media data, identify trends, and develop cognitive computing applications.	L3, L4	4 (a), 4 (b)	4
Apply supervised and unsupervised machine learning techniques, including classification, regression, clustering, and dimensionality reduction, using libraries and APIs to analyze data, evaluate models, and solve real-world problems.	L3, L4	4 (c), 5 (a), 5 (b), 5(c)	8
Implement deep learning models for applications like image classification, sentiment analysis, and reinforcement learning, while using big data technologies to process and visualize large datasets efficiently.	L3, L4		

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries an equal mark.

1. (a) Answer the following questions: 2

- i. "If it looks like a duck and quacks like a duck, it probably is a duck." What does this mean in Python's Duck Typing? How does it support Polymorphism? Give an example.

What happens if a user tries to access a private attribute (`_attribute`) directly? How can you retrieve the value of a private attribute without modifying the class definition?

- (b) A bank wants to restrict direct access to an account's balance. Write a Python class, **BankAccount**, that uses getter and setter methods to manage the balance securely. Ensure that the balance cannot be accessed or modified directly. The class should have:

- A constructor to initialize the balance.
- A getter and setter for the balance.
- Methods to deposit and withdraw money, with checks for valid amounts.

(c)

```
class Data:
    def __init__(self, value):
        self.value = value
```

```
x = Data(10)
y = x
y.value = 20
print(x.value)
```

What is the output of the above program? Why does modifying `y.value` also affect `x.value`?

2. (a) What is **recursion**? Implement a recursive solution to compute the n^{th} Fibonacci number and analyze its time complexity. 2
- (b) You are given a sorted list `[2, 5, 8, 12, 16, 23, 38, 56]`. Implement a recursive **binary search** function to locate the index of the element 23. If the element is not found, return -1. Furthermore, evaluate the time complexity and space complexity of your recursive binary search solution. 2
- (c) Given a list of integers, implement a Python program to sort the list using the **insertion sort** algorithm. For example, if the list is `[12, 11, 13, 5, 6]`, write the code to sort it and display the sorted result. Furthermore, analyze the best-case and worst-case time complexities of the insertion sort algorithm. 2

3. (a) Consider the following Python code:

```
from textblob import TextBlob
text = "Python is a wonderful programming language."
blob = TextBlob(text)
print(blob.tags)
```

What is the expected output of this code? Explain the significance of the output in the context of Parts-of-Speech tagging.

- (b) Write a Python program using **TextBlob** to classify the sentiment (positive, negative, or neutral) of a list of sentences. The program should:

Accept sentences as input. Perform sentiment analysis using TextBlob. Classify each sentence as positive, negative, or neutral based on the polarity score.

Test the program using the following sentences:

"I love the weather today!"

"The food was awful, and the service was terrible."

"It was an average experience."

- (c) Write a Python program that performs the following tasks using **TextBlob**: 2

- Take the following sentence:

"Running quickly towards the finishing line, the exhausted runners were hoping to win. There were different varieties of prizes."

- Tokenize the sentence into words.
- Perform stemming and lemmatization to return the base forms of each word.

Discuss which output (stemming or lemmatization) is more readable and explain why.


```

void Ball::bounceBall(RenderWindow
                      &window, Time &dt){
    float x = ballShape.getPosition().x;
    float y = ballShape.getPosition().y;
    x += dt.asSeconds() * m_DirectionX
                                * m_Speed;
    y += dt.asSeconds() * m_DirectionY
                                * m_Speed;
    ....
}

```

(b) Write a code snippet in SFML to move a bat left and right using keyboard input, while preventing it from going off-screen. [2]

(c) Write a code snippet for a basic collision detection between the ball and a bat using SFML bounding boxes. [2]

End of Questions

MID-SEMESTER EXAMINATION, April-2022

Game Programming with C++ (CSE 3545)

Programme: B.Tech.(CSE)
Full marks: 30

Semester: 6th
Time: 2 hours

Subject Learning Outcome	*Taxonomy Level	Question Number	Marks
Write, compile and execute simple C++ programs to solve computational problems.	L4, L1, L3	1a, 1b, 1c	2, 2, 2
Become familiar with basic games using SFML library	L4, L3, L4, L3, L3, L3, L4	2a, 2b, 2c, 3b, 3c, 5a, 5c	2, 2, 2 2, 2, 2 2
Implement Game Mechanics and Physics.	L4, L4, L3, L3, L3	3a, 4a, 4b, 4c, 5b	2, 2, 2 2, 2
Develop advanced game features like sprite sheets, vertex arrays, texture management, and layering views.			
Apply advanced object-oriented programming concepts like inheritance, polymorphism, and abstraction			
Implement file & socket I/O, sound effects, and refine the game's HUD & UI.			

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries an equal mark.

1. (a) Write a **swap** function that uses concept of pass by reference in C++. With a neat diagram explain how it differs from pass by value and pass by address. [2]

(b) Provide an explanation with an example highlighting the importance of parameterized constructor with default value in C++. [2]

(c) Determine and explain the output of the following code. [2]


```

class A {
public:
    A() { cout << "A "; }
    ~A() { cout << "~A "; }
};

void func() {
    static A obj;
}

int main() {
    func();
    func();
    cout << "Main ";
    return 0;
}

```

2. (a) Write C++ statements to open a window with 960 pixels wide by 540 pixels high. [2]
- (b) Find and explain the output of the following code. [2]

```

enum hello{
    a,b=99,c,d=-1
};

main(){
    enum hello m;
    cout<<a<<" "<<b<<" "<<c<<" "<<d<<endl;
    return 0;
}

```

- (c) Consider the two SFML classes, Texture and Sprite, that will take care of drawing sprites into the screen. Use the two classes to draw an image `soa.jpeg` onto the window of size 960×540 . [2]
3. (a) There are two sprites namely `spriteBee`, and `spriteBackground` loaded with proper texture and having size 25×25 and 800×600 respectively. However the

following game loop is not able display the `spriteBee`. State the reasoning behind it and modify the code for displaying it appropriately. [2]

```

while(window.isOpen()){
    window.clear();
    window.draw(spriteBee);
    window.draw(spriteBackground);
    window.display();
}

```

- (b) Write a code snippet for closing the window when Escape key is pressed. [2]
- (c) The `pollEvent` function of `RenderWindow` class puts data into the event object of the Event class that describes an operating system event. There would be many events stored in the queue. Construct a while loop to check for the event types `KeyPressed` and display a message, "**KeyPressed Detected**" on the std, if that event has occurred. [2]
4. (a) Create a program to generate a random 1-D array of size 10. Additionally shift each of the array elements 2 positions to right and filled the beginning two places with random elements from the range 20 to 30. Finally display both the arrays [2]
- (b) Write a code to display "Hello SOA" in the middle of the game window with a Text Object in "arial.ttf" font. [2]
- (c) The game has a Cloud Sprite `spriteCloud` which needs to move from left to right. Write a code for smooth transition of cloud from left to right. [2]
5. (a) The `Ball` class include a member function named `bounceBall`. The partial code for the member function is as follows. Complete the code that helps the ball to bounce form all sides of the window. [2]